Adjoint-Based Mesh Refinement

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Proposed Work

In this project, we propose to add adjoint capability to the existing Discontinuous Galerkin (DG) solver for output-based error estimation and mesh refinement. The goal is to use adjoints to determine which local discretization errors most strongly affect a scalar output of interest, such as the lift coefficient. Rather than refining the mesh only in regions where the flow appears complex, adjoint-based methods provide a systematic way to identify where refinement is most beneficial for improving a selected output.

Approach

We will begin by solving the DG problem on a coarse discretization, taken here to be $p = 1$, to obtain a converged coarse-space solution U_H. We will then solve the corresponding coarse adjoint problem to obtain ψ_H. After this, we will take the fine space to be a $p = 2$ discretization on the same mesh. The converged coarse solution U_H will be injected into the finer space to obtain U_h^H. Although the coarse residual is zero at convergence, the injected coarse solution does not in general satisfy the fine-space equations, so the fine-space residual

$$R_h(U_h^H)$$

is nonzero and can be used to estimate unresolved discretization error.

The output correction will be estimated using the adjoint-weighted residual

$$\delta J \approx -\psi_h^T R_h(U_h^H),$$

where ψ_h is the fine-space adjoint, calculated or approximated on the fine-space discretization. This correction can be localized to each element to define the error indicator

$$\epsilon_e = \left|\psi_{h,e}^T R_{h,e}(U_h^H)\right|.$$

Elements with larger values of ϵ_e will be identified as the most important candidates for refinement. The goal of the refinement study will be to improve the computed lift coefficient C_l to within a prescribed tolerance. Once the estimated error in C_l falls below a chosen tolerance (for example, 10^{-4}), the mesh adaptation process will be stopped.

Plan and Expected Results

The project tasking would proceed in three stages: in the first week, we will adapt the DG solver so that it can evaluate the coarse-space residual and Jacobian, solve the coarse adjoint problem, and evaluate the residual for a $p = 2$ discretization by injecting the coarse solution. In the second week, we will calculate or approximate the fine-space adjoint and construct the adjoint-weighted residual correction and element-wise error indicator. In the third week, we will compare adjoint-based refinement against uniform refinement at a comparable mesh size in order to assess whether adjoint-based refinement provides a more efficient improvement in output accuracy. By the end of the project, we expect to have implemented a discrete adjoint option within the DG solver that demonstrates local mesh refinement.